SPORTS ANALYTICS / QUANTITATIVE RESEARCH

EPA-Based Forecasting and Pricing

NFL · NBA · College Football

Mathematical framework, joint score distributions, moneyline and totals pricing, and production system design

Central objective
Use opponent-adjusted offensive and defensive efficiency, projected opportunity volume, and game-specific scoring simulation to estimate the joint predictive distribution

$$P(S_A = x, S_B = y \mid \mathcal{I}_t)$$

Derive projected scores, Over/Under probabilities, moneyline probabilities, fair odds, and expected value from the *same* underlying distribution.

01 / Modeling	02 / Pricing	03 / Implementation
EPA, PPP, PPA and team strength	Monte Carlo, fair odds and EV	Architecture, data and validation

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1 Executive framework and objectives

The aim is to forecast the probability of actual betting outcomes, not merely to rank teams by average efficiency. An EPA-based system must bridge three distinct layers: (i) team quality, (ii) opportunities to score, and (iii) the random distribution of final scores. For basketball, the analogous efficiency quantity is points per possession (PPP), optionally enriched by expected possession value (EPV). College football uses a separately calibrated EPA/PPA framework.

Let S_A and S_B denote *final* points scored, with the league's overtime and market-settlement rules incorporated. The fundamental object is

$$P(S_A = x, S_B = y \mid \mathcal{I}_t),$$

where $\mathcal{I}_t$ is the information actually available at forecast time t. From this one distribution:

$$\begin{aligned} E[T] &= E[S_A] + E[S_B], & T &= S_A + S_B, \\ E[M] &= E[S_A] - E[S_B], & M &= S_A - S_B, \\ P(\text{Over } L) &= P(T > L), & P(A \text{ wins}) &= P(M > 0). \end{aligned}$$

Modeling principle. EPA measures changes in expected value during a play; it is not directly an accurate final-score forecast. Pace, opponent adjustment, personnel, game context, and scoring randomness determine how efficiency becomes points.

1.1 Unified architecture

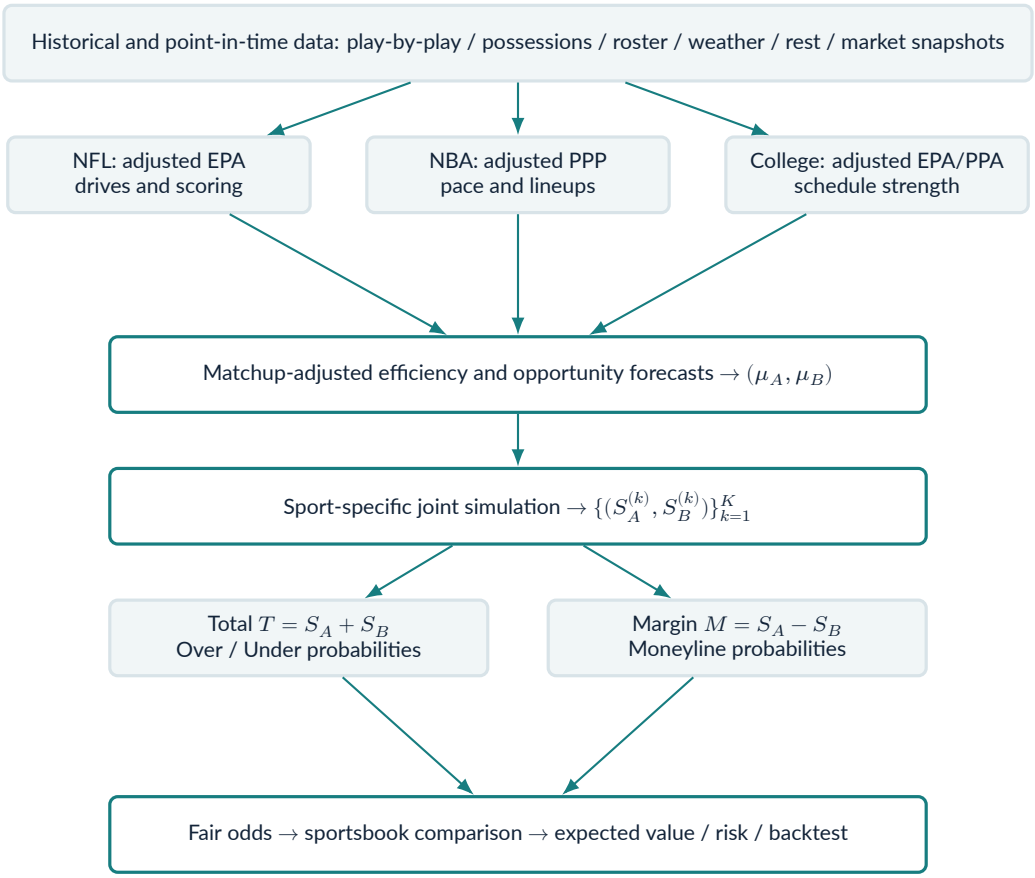


Figure 1: One common pricing system fed by three sport-specific statistical models.

2 Football EPA: the mathematical foundation

2.1 Expected points

Define a football game state $s = (d, y, x, \tau, q, ...)$, containing down, distance, field position, time remaining, period, and other relevant factors. Expected points is the conditional expectation of future *net* points from the current offensive perspective, under a clearly specified scoring horizon:

$$EP(s) = E[\text{future net points} \mid s] = \sum_{k \in \mathcal{K}} p_k(s) v_k.$$

Here $p_k(s)$ is the probability of a future signed scoring outcome k, and v_k is its point value. In practice, consistent handling of kickoffs, possession changes, and period boundaries is required.

2.2 Expected Points Added

For a standard play that does not immediately score or change possession, $EPA_i = EP(s_{i+1}) - EP(s_i)$. A more general signed-reward representation is

$$\boxed{EPA_i = r_i + \eta_i EP(s_{i+1}) - EP(s_i)}$$

r_i is the immediate signed scoring reward; $\eta_i = +1$ when the next state's expected points are measured from the original offense's perspective and -1 when measured from the opposing offense's perspective. Specific data providers implement their own scoring and boundary conventions.

Conversion example. If a team's expected points rise from 2.10 to 4.00 after converting third down, then $EPA = 4.00 - 2.10 = +1.90$.

Interception example. Suppose the offense had $EP = 2.10$, throws an interception, and the opponent starts with $EP = 3.00$. With no immediate score and values expressed from the original offense's perspective, $EPA = -3.00 - 2.10 = -5.10$. This explains why EPA captures the value of turnovers more meaningfully than raw yardage.

2.3 Offensive and defensive team EPA

For a team j:

$$OEPA_j = \frac{\sum_{i=1}^{n_j} EPA_i^{off}}{n_j}, \qquad DEPA_j = \frac{\sum_{i=1}^{m_j} EPA_i^{allowed}}{m_j}.$$

With $DEPA$ defined as *EPA allowed*, a smaller or more negative value represents stronger defense.

Illustrative team	Offensive EPA/play	Defensive EPA allowed/play
Team A	$+0.180$	-0.070
Team B	$+0.025$	$+0.085$

Table 1: Team A is stronger on both raw measures, but opponent adjustment remains necessary.

3 Estimating latent team strength

Raw EPA is confounded by schedule quality, recency, injuries, and game context. Model play-level EPA e_i using offense and defense effects:

$$e_i = \alpha + O_{a(i)} + D_{b(i)} + \beta^{\mathsf{T}} X_i + \epsilon_i$$

$O_{a(i)}$ denotes the offensive strength of the team running play i; $D_{b(i)}$ denotes the *defensive weakness* of its opponent (positive means weaker defense). Identification constraints include $\sum_j O_j = \sum_j D_j = 0$.

3.1 Regularization and early-season uncertainty

To prevent extreme estimates from small samples, estimate θ with weighted ridge regression:

$$\hat{\theta} = \arg\min_{\theta} \left\{ \sum_i w_i (e_i - \hat{e}_i)^2 + \lambda_O \sum_j O_j^2 + \lambda_D \sum_j D_j^2 \right\}.$$

Shrinkage is especially important in college football, where team schedules and information quality vary considerably. An approximate Bayesian-style shrinkage formulation is

$$\hat{O}_j = r_j O_j^{sample} + (1 - r_j) O_j^{prior}, \qquad r_j = \frac{n_{eff,j}}{n_{eff,j} + k}.$$

Tune k on earlier seasons only; do not select it to improve the eventual test set.

3.2 Recency weighting and changing team quality

Use $w_i = e^{-\lambda \Delta t_i}$, where Δt_i is the age of the observation, giving

$$EPA_{weighted} = \frac{\sum_i w_i EPA_i}{\sum_i w_i}.$$

For a more principled dynamic model, use $O_{j,t} = O_{j,t-1} + u_{j,t}$, where the latent innovation $u_{j,t}$ reflects changing team strength. A state-space or hierarchical Bayesian model can explicitly represent posterior uncertainty.

3.3 Matchup and play-type features

Do not aggregate passing, rushing, early-down efficiency, explosive plays, success rate, and red-zone performance prematurely. For play type k:

$$\hat{e}_{A,k} = \alpha_k + O_{A,k} + D_{B,k} + \beta_k^{\mathsf{T}} X_{AB}, \qquad \delta_A = \sum_k \pi_{A,k} \hat{e}_{A,k}.$$

$\pi_{A,k}$ is the *pregame predicted* play-selection share. Using the actual play mix from the future game would introduce leakage. Projected quarterbacks and likely playing time must also be reflected in the feature snapshot.

4 NFL: converting EPA into final-score forecasts

The fundamental decomposition is *efficiency* $\times$ *opportunity*. First estimate how well each offense will perform against the opposing defense; then estimate its number and quality of opportunities. Treat quarterback availability, playing style, weather, field position, and game state as forecast inputs.

4.1 Matchup-adjusted efficiency and volume

$$\delta_A = f(O_A, D_B, QB_A, \text{matchup}, \text{weather}), \qquad \delta_B = f(O_B, D_A, QB_B, \text{matchup}, \text{weather}).$$

Let N_A denote projected offensive drives and L_A denote projected offensive plays per drive. In general, $E[N_A L_A]$ must account for covariance; simply multiplying separate averages may be an approximation.

$$\hat{N} = f_\theta(Pace_A, Pace_B, PassRate_A, PassRate_B, Weather, Context).$$

Successful offenses can extend drives; rush rate changes clock usage; turnovers create different field positions; score differentials change play calling. Historical play counts alone do not fully describe game tempo.

4.2 A parsimonious scoring bridge

For a first-stage regression, let b_A be the neutral expected *offensive* points per drive given projected starting field position, δ_A the team's EPA advantage relative to that baseline, and κ an empirically calibrated conversion parameter:

$$\boxed{\mu_A^{off} = N_A b_A + \kappa N_A L_A \delta_A}, \qquad \boxed{\mu_B^{off} = N_B b_B + \kappa N_B L_B \delta_B}.$$

This is a forecast specification, *not* an EPA accounting identity. The neutral baseline and efficiency term must be fitted consistently to avoid double counting; defensive scores, safeties, special teams, turnovers and overtime require separate treatment.

A more flexible alternative is $\mu_A = g_\theta(\delta_A, N_A, L_A, FieldPosition_A, RedZone_A, SpecialTeams_A, X_{AB})$, trained directly against available historical outcomes.

4.3 Preferred advanced approach: drive-outcome modeling

For each drive starting in state s, learn $P(Y = k \mid s, \text{matchup})$ across mutually exclusive outcomes. A model might use multinomial logistic regression,

$$P(Y = k \mid X) = \frac{\exp(f_k(X))}{\sum_j \exp(f_j(X))},$$

or a calibrated gradient-boosted classifier. An example outcome vector (*illustrative only*) is:

Drive outcome	Probability
Offensive touchdown	0.25
Field goal	0.19
No score	0.51
Immediate defensive touchdown	0.01
Other explicitly modeled outcomes	0.04
Total	1.00

Under a simplified seven-point touchdown convention, the expected points from the first two offensive scoring categories are $7(0.25)+3(0.19)$; other outcomes and actual conversion choices are necessary for a complete calculation. A defensive touchdown is assigned to the *opponent's* scoreboard. The transition model must also update game clock, field position, possession, and score.

Why drive simulation improves forecasts. EPA measures incremental expected value, while a drive model converts team quality into a valid sequence of scoring and non-scoring events. This preserves the discrete nature of football scores and supports multiple betting lines with one generated score distribution.

5 College football: EPA with stronger schedule adjustment

The same game-state logic applies, but collegiate scheduling, roster turnover, talent disparity, tempo, garbage time, and overtime require separate model calibration. CollegeFootballData provides PPA and adjusted efficiency metrics; PPA and nflverse EPA are related valuation concepts, *not* numerically interchangeable data series.

5.1 Opponent-adjusted prediction

For Team A against Team B,

$$OEPA_A^{adj} = \alpha + O_A, \qquad \boxed{EPA_{A\to B} = \alpha + O_A + D_B + \beta^{\mathsf{T}} X_{AB}}.$$

Repeat for the reverse matchup. Shrink estimates from scarce observations toward priors informed by previous performance, roster continuity, transfers, coaching changes, and other features that demonstrate out-of-sample value. Do not treat dominant performances against weak opponents as unconditional evidence of elite efficiency.

Component	NFL	College football
Efficiency	NFL-calibrated EPA/play	College-calibrated EPA or PPA
Schedule adjustment	Important	Particularly important
Early-season prior	Helpful	Often essential
Tempo	Drives, clock, play mix	Wider scheme variation
Garbage time	Model explicitly	More consequential for rating noise
Overtime	NFL-specific	NCAA-specific

5.2 Garbage time and market rules

Downweight plays in effectively decided games *when estimating latent team quality*, if validation supports it; nevertheless simulate those late-game points when forecasting full-game totals. Use the NCAA's relevant overtime procedures, not an NFL overtime approximation. Market definitions (including whether overtime counts) must match the specific quoted wager.

6 NBA: use possession efficiency and expected possession value

An NBA forecast is naturally built around *offensive possessions*, not NFL-style plays. The practical pregame baseline is adjusted points per possession (PPP); a more advanced event-level model can use Expected Possession Value (EPV).

6.1 Expected possession value

Let $\mathcal{F}_t$ contain all information observed during a current possession and let R be the remaining points generated during it:

$$V(s_t) = E[R \mid \mathcal{F}_t].$$

If a pass increases EPV from 1.08 to 1.42, its incremental expected-value improvement is $+0.34$ points, subject to a consistent scoring and reward convention. EPV may use ball/player positions, shot clock, defender locations, and possession history. See Cervone et al. in the references for the underlying basketball modeling concept.

6.2 A tractable adjusted-PPP model

For Team A, define offensive and defensive ratings as points per 100 possessions:

$$ORtg_A = 100\frac{PTS_A}{Poss_A^{off}}, \quad DRtg_A = 100\frac{PTSAllowed_A}{Poss_A^{def}}, \quad OPPP_A = \frac{ORtg_A}{100},\ DPPP_A = \frac{DRtg_A}{100}.$$

Let ℓ denote league-average PPP. A simple additive opponent adjustment is

$$\boxed{\widehat{PPP}_{A\to B} = \ell + (OPPP_A - \ell) + (DPPP_B - \ell) = OPPP_A + DPPP_B - \ell}.$$

Example: $OPPP_A = 1.18$, $DPPP_B = 1.14$, and $\ell = 1.12$ imply $\widehat{PPP}_{A\to B} = 1.20$.

This is a baseline, not a replacement for opponent-adjusted, lineup-aware estimation. In addition, offensive rebounds typically extend an existing possession rather than automatically creating a new one. Consistent possession definitions are necessary.

6.3 Project pace and points

Let P_A, P_B denote historically adjusted pace. A simple baseline is $\hat{N} = (P_A + P_B)/2$. A richer forecast uses

$$\hat{N} = \alpha + \beta_1 P_A + \beta_2 P_B + \beta_3 H + \beta_4 R + \beta_5 L + \epsilon,$$

where H represents home-court factors, R rest and schedule, and L expected lineups. The teams experience a substantially *shared* possession count, rather than independent pace processes.

$$\boxed{\mu_A^{reg} \approx \hat{N} \cdot \widehat{PPP}_{A\to B}}, \qquad \boxed{\mu_B^{reg} \approx \hat{N} \cdot \widehat{PPP}_{B\to A}}.$$

Generally, $E[N \times PPP]$ includes a covariance term. The product of two point forecasts is a useful approximation; the simulator should capture their dependence. These equations are for *regulation*; full-game probabilities need overtime.

Worked NBA scoring example. Suppose $\hat{N} = 99$, $\widehat{PPP}_A = 1.18$, and $\widehat{PPP}_B = 1.12$. Then

$$\mu_A^{reg} = 116.82, \quad \mu_B^{reg} = 110.88, \quad \mu_T^{reg} = 227.70, \quad \mu_M^{reg} = +5.94.$$

The approximate projected score is 117–111. Full-game pricing must account for overtime and game-end dynamics.

6.4 Possession outcome simulation

For an advanced system, learn the conditional distribution of points for a possession R_i:

$$P(R_i = k \mid A, B, X_i), \qquad E[R_i] = \sum_k kP(R_i = k), \qquad S_A^{off} = \sum_{i=1}^{N_A} R_i.$$

Include zero, one, two, three, four and upper-tail outcomes as appropriate; unusual scoring events (e.g., technical fouls) need explicit accounting. Important features include turnover propensity, offensive rebounds, free throws, lineup combinations, and shot selection. Overtime and end-of-game intentional fouling should be modeled rather than handled by adding arbitrary independent score noise.

7 Joint score distribution: the bridge to markets

For both moneyline and totals, forecast the two team scores *jointly*. A useful pedagogical approximation is a bivariate normal distribution:

$$\begin{bmatrix} S_A \\ S_B \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_A \\ \mu_B \end{bmatrix}, \begin{bmatrix} \sigma_A^2 & \rho\sigma_A\sigma_B \\ \rho\sigma_A\sigma_B & \sigma_B^2 \end{bmatrix} \right).$$

This convenient approximation allows negative and non-integer scores and is therefore not recommended as the final production distribution, especially for football.

7.1 Totals

For $T = S_A + S_B$, the expectation and variance are

$$\boxed{\mu_T = \mu_A + \mu_B}, \qquad \boxed{\sigma_T^2 = \sigma_A^2 + \sigma_B^2 + 2\rho\sigma_A\sigma_B}.$$

Under the normal approximation, with standard normal CDF Φ,

$$\boxed{P(\text{Over } L) \approx 1 - \Phi\left(\frac{L - \mu_T}{\sigma_T}\right)}.$$

For an integer total, compute the push mass $P(T = L)$ explicitly using a discrete score distribution and apply the sportsbook's settlement rules.

7.2 Moneyline

For $M = S_A - S_B$,

$$\boxed{\mu_M = \mu_A - \mu_B}, \qquad \boxed{\sigma_M^2 = \sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}.$$

$$\boxed{P(A \text{ wins}) \approx \Phi\left(\frac{\mu_M}{\sigma_M}\right)}.$$

Score correlation enters the total variance and margin variance with *opposite signs*. This is an important reason to use a joint model. The normal formula does not fully resolve NFL ties or overtime and must be treated as an approximation.

8 Worked forecasts: NFL and NBA

8.1 NFL example: Team A versus Team B

All numbers in this example are hypothetical. Let $\mu_A = 24.80$, $\mu_B = 21.50$, $\sigma_A = 10$, $\sigma_B = 9$, and $\rho = 0.20$.

Model parameter	Team A	Team B
Expected score	24.80	21.50
Score standard deviation	10.00	9.00

Table 2: Illustrative parameters; shared score correlation is 0.20.

Expected total and margin: $\mu_T = 46.30$, $\mu_M = +3.30$. **Margin uncertainty:**

$$\sigma_M^2 = 100 + 81 - 2(0.20)(10)(9) = 145, \qquad \sigma_M = 12.041595.$$

Consequently,

$$P(A \text{ wins}) \approx \Phi(3.3/12.041595) \approx \boxed{60.80\%}.$$

Total uncertainty:

$$\sigma_T^2 = 100 + 81 + 2(0.20)(10)(9) = 217, \qquad \sigma_T = 14.730920.$$

At a sportsbook total of 44.5:

$$P(T > 44.5) \approx 1 - \Phi((44.5 - 46.3)/14.730920) \approx \boxed{54.86\%}.$$

The Under probability is about 45.14%. The modeled total exceeds the quoted line by 1.8 points, but that alone is not a probability or an indication of positive EV.

8.2 NBA example: Team A versus Team B

Using the regulation projection from Section 6, $\mu_A = 116.82$, $\mu_B = 110.88$, $\mu_T = 227.70$, and $\mu_M = +5.94$. Assume *illustrative* standard deviations $\sigma_M = 12$ and $\sigma_T = 16$:

$$P(A \text{ wins in the approximate regulation model}) \approx \Phi(5.94/12) \approx \boxed{68.97\%},$$

$$P(T > 224.5 \text{ in regulation}) \approx 1 - \Phi((224.5 - 227.7)/16) \approx \boxed{57.93\%}.$$

These are *regulation approximations*, not correctly settled full-game sportsbook prices. The production engine must simulate overtime.

8.3 NBA shared-pace uncertainty

With random common possessions N and temporarily fixed rates r_A, r_B, suppose $T \approx N(r_A + r_B) + \epsilon_T$ with conditional residual variance. The law of total variance gives

$$\boxed{\mathrm{Var}(T) = (r_A + r_B)^2 \mathrm{Var}(N) + E[\mathrm{Var}(T \mid N)]}.$$

The first term is pace uncertainty; the second is scoring uncertainty conditional on pace. This separation is valuable for pricing NBA totals.

9 Monte Carlo as the production pricing engine

Use sport-specific simulation to generate K internally consistent *final* score pairs $\{(S_A^{(k)}, S_B^{(k)})\}_{k=1}^{K}$. A practical initial setting is $K = 100{,}000$. The two scores should share latent pace, game environment and state transitions; do not draw them independently unless that assumption is justified.

Layer	NFL / college football	NBA
Opportunity	Drives and plays	Offensive possessions
Efficiency	Adjusted matchup EPA	Adjusted matchup PPP
State transitions	Possession, field position, clock, score	Possession, lineup, shot clock, score
Outcomes	TD, FG, punt, turnover, safety, etc.	Possession point outcomes, turnovers, etc.
Endgame	League-specific late-game / overtime	Fouling and NBA overtime
Output	Joint final-score samples	Joint final-score samples

For football, simulate $s_{d+1} \sim P(s_{d+1} \mid s_d, X_{AB})$. For basketball, draw a common latent game environment $Z^{(k)} \sim P(Z \mid \mathcal{J}_t)$, draw $N^{(k)} \sim P(N \mid Z^{(k)})$, then draw possession outcomes conditional on the same environment, teams, and lineups.

9.1 Pricing all markets from the simulated samples

$$\widehat{P}(A \text{ wins}) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{S_A^{(k)} > S_B^{(k)}\},$$

$$\widehat{P}(\text{Over } L) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{S_A^{(k)} + S_B^{(k)} > L\},$$

$$\widehat{P}(\text{Push at } L) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{S_A^{(k)} + S_B^{(k)} = L\}.$$

The same sample set can price alternative totals, spreads, and main lines without refitting a different statistical model. Model spread results from $M + h$ where h is the home betting handicap (e.g., $h = -3.5$ means home -3.5).

9.2 Reusable Python pricing function

Listing 1: Market probabilities given final-score simulation arrays.

```python
import numpy as np

def price_markets(home_scores, away_scores,
                  total_line, spread_line=0.0):
    """spread_line is the home handicap (-3.5 for home -3.5)."""
    home = np.asarray(home_scores)
    away = np.asarray(away_scores)
    assert home.shape == away.shape and len(home) > 0

    total = home + away
    margin = home - away
    adjusted_margin = margin + spread_line

    return {
        "expected_home": float(np.mean(home)),
        "expected_away": float(np.mean(away)),
        "expected_total": float(np.mean(total)),
        "expected_margin": float(np.mean(margin)),
        "home_win": float(np.mean(margin > 0)),
        "away_win": float(np.mean(margin < 0)),
        "tie": float(np.mean(margin == 0)),
        "over": float(np.mean(total > total_line)),
        "under": float(np.mean(total < total_line)),
        "total_push": float(np.mean(total == total_line)),
        "home_cover": float(np.mean(adjusted_margin > 0)),
        "away_cover": float(np.mean(adjusted_margin < 0)),
        "spread_push": float(np.mean(adjusted_margin == 0)),
    }
```

NFL tie settlement requires market-specific postprocessing; NBA and college simulations should play out applicable overtime. At $K = 100{,}000$ *independent* samples, the Monte Carlo standard error for a probability of 50% is approximately 0.16 percentage points. This is not the larger error caused by estimated parameters or model misspecification.

10 Fair odds, implied probability, and betting expected value

A model-estimated probability is a prerequisite, not a sufficient condition, for a value signal. Compare it with the *available* payout and settlement conditions.

10.1 American odds and bookmaker margin

If $a > 0$ denotes the magnitude of American odds,

$$P_{implied}(-a) = \frac{a}{a+100}, \qquad P_{implied}(+a) = \frac{100}{a+100}.$$

Examples: $-110 \mapsto 52.3810\%$, $-150 \mapsto 60.0000\%$, $+130 \mapsto 43.4783\%$. These are *break-even* probabilities for quoted prices, not necessarily fair outcome probabilities. For a standard binary market, a simple proportional no-vig approximation is $p_A^{fair} = q_A/(q_A + q_B)$, where q_A, q_B are the two raw implied probabilities. Other de-vig methods produce different estimates.

10.2 Expected value and illustrative NFL prices

If p_w is the probability of a winning settlement, p_l of a losing settlement, and b the net profit on each dollar staked on a win, with pushes returning stake:

$$\boxed{EV = p_w b - p_l}.$$

Consider illustrative Team A win probability 0.608 and a sportsbook offer of -135. Then $b = 100/135 = 0.740741$ and

$$EV \approx 0.608(0.740741) - 0.392 = \boxed{+5.84\%}.$$

This ignores tie-settlement details in the pedagogical approximation. A 60.8% win probability implies decimal fair odds $1/0.608 \approx 1.644737$, or approximately American -155. The -135 price breaks even at $135/235 = 57.4468\%$.

For the illustrated NFL Over 44.5 at -110, using $P(Over) = 0.548627$, the break-even probability is 0.523810 and

$$EV \approx 0.548627(100/110) - 0.451373 \approx \boxed{+4.74\%}.$$

Hypothetical selection	Model win probability	Price break-even	Model EV
Team A moneyline (-135)	60.80%	57.45%	$+5.84\%$
Over 44.5 (-110)	54.86%	52.38%	$+4.74\%$

Table 3: Illustrations only: no live sportsbook prices or recommendations.

10.3 Uncertainty-aware signals

Marginal positive EV should not automatically trigger a bet. Let $\hat{p}$ be a point probability forecast and u_p a calibrated forecast-uncertainty adjustment. Define

$$p_{conservative} = \hat{p} - u_p.$$

Apply a prespecified minimum-EV threshold using this conservative estimate. Choose the adjustment on historical training/validation data; do not reverse-engineer it to justify an individual opportunity. Correlated bets within a game should also share exposure limits.

11 System architecture and data engineering

Use a Python analytical layer, a relational database for normalized historical and point-in-time records, Parquet files for large play-by-play collections, and a clear distinction between sports observations, model forecasts, and sportsbook price snapshots.

11.1 Technology stack

Subsystem	Suggested technology / responsibility
Ingestion	Python clients, provider APIs, scheduled downloads
Raw files	Parquet, versioned immutable source extracts
Relational storage	PostgreSQL or SQL Server
Transformations	pandas / NumPy / Polars, validated feature pipelines
Statistical estimation	scikit-learn, statsmodels; XGBoost or PyMC as needed
Simulation	NumPy / SciPy; efficient vectorization
Serving	FastAPI with versioned forecast endpoints
Presentation	Plotly Dash for forecasts and market comparison
Deployment / jobs	Docker and orchestration (e.g., Prefect / Airflow)

11.2 Recommended data sources

- **NFL:** nflverse / nflreadpy for historical NFL play-by-play and EPA-related fields; check provider documentation and data availability.
- **College football:** CollegeFootballData for play-by-play, PPA, team metrics and schedules, subject to the applicable API tier and rights.
- **NBA:** NBA Stats for team ratings, pace and statistical definitions; nba_api and pbpstats can facilitate accessible endpoint retrieval and possession parsing. Verify current interfaces, limits, and licensing before production.
- **Sportsbook prices:** Choose an authorized odds source and verify actual sport, bookmaker and alternate-line coverage. A feed offering standard main lines may not offer every FanDuel alternate total or spread. Capture quotes historically and with timestamps.

11.3 Relational data model

Listing 2: Recommended logical database entities; illustrated field names.

```
games(game_id, sport, season, scheduled_start,
      home_team_id, away_team_id, status)

play_by_play(game_id, play_id, event_timestamp,
             offense_team_id, defense_team_id,
             play_type, epa, game_state)

team_feature_snapshots(team_id, sport, snapshot_timestamp,
                       adjusted_offensive_rating,
                       adjusted_defensive_rating, pace,
                       availability_adjustments, model_version)

forecast_snapshots(forecast_id, game_id, generated_at,
                   information_cutoff, model_version,
                   expected_home_score, expected_away_score,
                   simulation_reference)

sportsbook_odds(game_id, sportsbook, market_type,
                market_scope, selection, line, american_odds,
                captured_at, provider_market_id)

market_predictions(forecast_id, sportsbook_market_id,
                   model_probability, implied_probability,
                   expected_value, pricing_timestamp)
```

Store full simulation outputs as compressed files and retain a reference in the forecast database. Include explicit market scope so full-game totals, first-half totals and regulation-only markets cannot be matched accidentally. Prices and predictions require their own *captured_at* / *information_cutoff* columns to support valid retrospective comparisons.

11.4 Ten-stage processing workflow

#	Pipeline stage	Output
1	Ingest observations	Raw, source-stamped data
2	Validate and normalize	Consistent event and possession tables
3	Engineer features	EPA, PPP, matchup, pace, lineup features
4	Estimate team strengths	Adjusted offense / defense ratings
5	Produce pregame forecast	Efficiency and opportunity distributions
6	Simulate final games	Joint score sample collection
7	Price relevant markets	Probabilities, push mass, fair odds
8	Match sportsbook quote	Timestamp-consistent price comparison
9	Apply risk / uncertainty rules	Qualified model signals
10	Evaluate	Calibration, error, market benchmark, realized results

Leakage control: a forecast issued 24 hours before the game cannot use a starting-lineup announcement made two hours before the game. Join every source as of its real historical availability timestamp, not merely by game date.

12 Backtesting and model validation

A model with a high correlation between estimated and observed scores is not necessarily well calibrated, and a calibrated model is not automatically profitable after transaction costs and sportsbook margin. Evaluate three separate objectives: score accuracy, probability quality, and price-relative performance.

12.1 Chronological walk-forward design

Training observations	Held-out test season
Through 2022	2023
Through 2023	2024
Through 2024	2025
Through 2025	2026

Refit within each historical information cutoff. Use training-period-only tuning, separate same-day versus earlier forecasts, and preserve raw odds snapshots captured before game start. Avoid random game splits that leak future team states or roster changes into past estimates.

12.2 Score error and distribution quality

For forecast and realized total $(\hat{T}_i, T_i)$ or margin $(\hat{M}_i, M_i)$,

$$RMSE_T = \sqrt{\frac{1}{n}\sum_i (\hat{T}_i - T_i)^2}, \qquad RMSE_M = \sqrt{\frac{1}{n}\sum_i (\hat{M}_i - M_i)^2}.$$

RMSE evaluates the predicted means, not the variance or tail behavior. Evaluate full predictive distributions with proper scoring rules such as CRPS or discrete log scores, plus prediction-interval coverage.

12.3 Moneyline probability calibration

For win outcome $y_i \in \{0, 1\}$ and forecast p_i (with ties handled under a declared convention):

$$\boxed{BS = \frac{1}{n}\sum_i (p_i - y_i)^2}, \qquad \boxed{LL = -\frac{1}{n}\sum_i \{y_i \log p_i + (1 - y_i)\log(1 - p_i)\}}.$$

Lower Brier score and log loss are better. Check reliability curves: events assigned 60% probability should win about 60% of the time across representative, sufficiently large samples.

12.4 Total-line calibration and consistency

Evaluate Over probabilities at *actual quoted thresholds* rather than just comparing a predicted total with a realized total. A coherent score distribution guarantees monotonicity:

$$P(T > 45.5) \leq P(T > 44.5).$$

Independent models for every alternate total can violate this constraint, whereas one underlying simulated distribution cannot.

12.5 Market benchmarking

Compare against (i) simple historical scoring baselines, (ii) opponent-adjusted EPA/PPP, and (iii) sportsbook consensus after a stated de-vig procedure. Optionally fit a calibrated market-plus-model ensemble, but retain the EPA-only signal to test independent predictive information. The sportsbook price is not an independent observation of ground truth.

Track closing-line value, realized ROI, calibration, error distributions and exposure-adjusted returns. Neither a short run of positive ROI nor favorable closing-line value alone proves a durable edge. Multiple bets from the same game share event risk.

13 Advanced extensions

Enhancement	Mathematical tool	Primary reason
Time-varying ratings	Bayesian state-space model	Respond to genuine team-strength changes
Schedule adjustment	Hierarchical regression	Reduce opponent-composition bias
Tactical matchups	Multivariate nonlinear model	Capture scheme interactions
Shared game environment	Latent-variable model	Preserve total / margin correlation
Personnel availability	Player / lineup contribution model	Adjust for injuries and substitution
Distribution tails	Mixture / conditional event model	Better price unusual outcomes
Forecast uncertainty	Posterior predictive simulation	Avoid unwarranted confidence
Market information	Validated model ensemble	Test additional signal beyond odds

13.1 Jointly model pace and efficiency

Instead of independently sampling scoring rate and opportunity count, introduce common latent conditions Z:

$$Z \sim P(Z \mid X), \quad N \sim P(N \mid Z, X), \quad PPP_A \sim P(PPP_A \mid Z, X), \quad PPP_B \sim P(PPP_B \mid Z, X).$$

Adapt the same factorization to football using drive volume and efficiency. This allows a scenario with unusual tempo to affect both offenses simultaneously.

13.2 Distinguish parameter uncertainty and game randomness

Instead of using a fixed estimate $\hat{O}_A = 0.15$, simulate team strength from its posterior $O_A \mid \mathcal{D} \sim P(O_A \mid \mathcal{D})$, then simulate game outcomes conditional on the sampled parameters. The law of total variance states

$$\boxed{\mathsf{Var}(S \mid \mathcal{D}) = E[\mathsf{Var}(S \mid \theta, \mathcal{D}) \mid \mathcal{D}] + \mathsf{Var}(E[S \mid \theta, \mathcal{D}] \mid \mathcal{D}).}$$

The first term is game-outcome uncertainty conditional on parameters; the second is parameter-induced predictive uncertainty. This distinction is particularly useful early in a season and in data-sparse collegiate matchups.

14 Recommended implementation roadmap

Phase	Build	Deliverable / validation checkpoint
I: Baseline	Separate adjusted EPA/PPP models, pace forecasts and empirically calibrated score distributions	Projected scores, fair ML / O&U prices, genuine walk-forward backtests
II: Simulation	Football drive-level and NBA possession-level engines, shared environment, lineups, league overtime	Internally coherent joint score simulations and better out-of-sample calibration
III: Live analytics	Timestamped odds feed, alternate markets, EV thresholds, dashboards and monitored exposure	Repeatable market comparison, risk monitoring and model diagnostics

14.1 Suggested dashboard views

Display each matchup's adjusted offensive/defensive ratings, projected opportunity volume, expected score and margin, score/total distributions, ML and alternate-line probabilities, fair odds and no-vig consensus, and point-in-time available sportsbook quotes. Include forecast version, injury/lineup assumptions, uncertainty intervals, and links to historical calibration and realized outcomes. Clearly identify which markets include overtime.

15 Final synthesis

The three leagues should share one distribution-to-market pricing API but have separately fitted sport-specific statistical and simulation models:

Layer	NFL	NBA	College football
Efficiency	EPA/play	PPP and optional EPV	College EPA/PPA
Volume	Drives / plays	Possessions	Drives / plays
Strength	Dynamic opponent-adjusted	Adjusted offensive / defensive	Hierarchical opponent-adjusted
Simulation	Drive-state transitions	Possession-state transitions	College drive transitions
Pricing	One joint-score engine for totals, moneyline, spreads, pushes and odds comparison		

The complete research target can be written as a posterior predictive mixture:

$$P(S_A, S_B \mid \mathcal{I}_t) = \int P(S_A, S_B \mid \theta, N, X)\, P(\theta, N \mid \mathcal{I}_t)\, d\theta\, dN.$$

θ captures underlying efficiency and team-strength parameters; N is scoring opportunity volume; X denotes relevant game conditions. In practice, draw parameters, opportunity counts and game outcomes, then price every market from the same simulated final-score pairs.

The most valuable early investment is **well-adjusted team strength and honest probability calibration**. A sophisticated simulator cannot compensate for systematically biased offensive / defensive ratings.

References and implementation resources

[1] **nflverse / nflreadpy.** Data access and documentation for NFL play-by-play and EPA-related data. https://nflreadpy.nflverse.com/ and https://nflverse.nflverse.com/.

[2] **nflfastR.** NFL play-by-play and expected-points modeling documentation. https://www.nflfastr.com/.

[3] **CollegeFootballData.** API and college football PPA, play-by-play, schedule and team analytics. https://collegefootballdata.com/.

[4] **NBA Stats.** Official statistical glossary and team metrics, including offensive rating, defensive rating, pace and possessions. https://www.nba.com/stats/help/glossary.

[5] **nba_api.** Python interfaces for NBA data endpoints. https://github.com/swar/nba_api.

[6] **pbpstats.** Play-by-play and possession parsing tools. https://github.com/dblackrun/pbpstats.

[7] **Cervone et al.** Work on Expected Possession Value and basketball possession-outcome prediction; useful conceptual grounding for advanced EPV modeling.

[8] **Brier (1950).** "Verification of forecasts expressed in terms of probability." Foundation for the Brier score.

[9] **Gneiting and Raftery (2007).** "Strictly Proper Scoring Rules, Prediction, and Estimation." Framework for probabilistic forecast evaluation.

Usage note. Worked games, score variances and sportsbook offers in this report are illustrative. Nothing here represents an actual contemporaneous game forecast, available sportsbook price, guaranteed return, or investment recommendation. Provider coverage, data terms, league rules, and specific market settlement must be verified at deployment time.